

- 3 Sickle cell anaemia is a disease caused by a mutation in the *HBB* gene.

The presence of the mutant *HBB* allele in the genome of a person can be detected using a variety of techniques.

- (a) DNA can be extracted and purified from cell samples taken from a person. The polymerase chain reaction (PCR) can be used to amplify the region of DNA containing the mutant *HBB* allele, if it is present in the genome of the person.

This PCR process amplifies only the mutant *HBB* allele and not other DNA sequences.

Outline the steps involved in PCR **and** explain how PCR can be carried out so that **only** the mutant *HBB* allele is amplified.

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- (b) Restriction enzymes and electrophoresis can be used to determine the alleles of the *HBB* gene that make up the genotype of a person.

The restriction enzymes:

- produce one fragment when only the mutant *HBB* allele is present
- produce two smaller fragments when only the normal *HBB* allele is present.

Figure 3.1 shows the results of electrophoresis of DNA samples from four people, 1 to 4. Only DNA fragments from the *HBB* gene locus are shown.

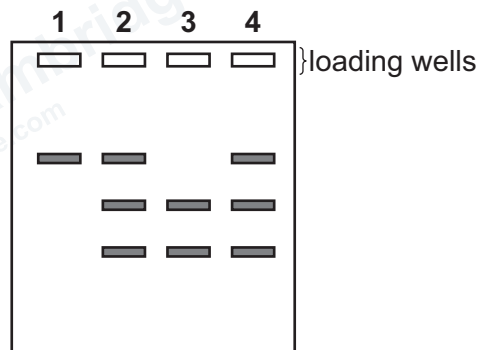


Figure 3.1

State which people, 1–4, show:

a genotype that is homozygous for the normal *HBB* allele

a genotype that is homozygous for the mutant *HBB* allele

a genotype that is heterozygous for the *HBB* gene

[2]



- (c) Microarray analysis can be used to detect the mutant *HBB* allele in the genome of a person.

Outline how microarray analysis is used to detect the presence of the mutant *HBB* allele in the genome of a person.

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